Chinyere Ukpong

Full-stack TypeScript developer (React, Node.js, MongoDB) building AI-powered products
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EDUCATION

BSc. Applied Computer Technology (Major in Software Engineering) | United States International University-Africa — Jan 2021 - Sept 2025

SKILLS SUMMARY

- **Programming & Markup:** JavaScript (ES6+), TypeScript, HTML5, CSS3
- **Frontend Frameworks & UI:** Next.js, React.js, Three.js, React Native, Tailwind CSS, GSAP
- **Backend & APIs:** Node.js, Express.js, RESTful API Design, WebSockets (Socket.IO), JWT Authentication & Authorization, OAuth 2.0, Zod, Rate Limiting
- **AI & Queues:** Gemini API (LLM integration), prompt design, BullMQ, Redis (Upstash)
- **Databases & Tools:** AWS EC2, systemd, Caddy, MongoDB (Mongoose), PostgreSQL, Docker, Appwrite, Git/GitHub, Vite, Render, Vercel, Cloudinary, VS Code, WebStorm
- **Testing:** Vitest, Jest, Supertest

WORK EXPERIENCE

Frontend Developer | Aerosub Solutions Ltd. — March 2026 - Current

- Built and maintained features across a multi-tenant SaaS inspection platform using Next.js 16, React 19, and TypeScript with nested dynamic routing.
- Developed interactive 2D/3D and 360° panorama viewers with defect positioning using Three.js and Canvas-based rendering.
- Integrated REST APIs throughout the application using Axios, handling authentication, query scoping, error states, and silent token refresh.
- Implemented file upload pipelines using presigned S3 URLs with concurrent workers, real-time progress tracking, and drag-and-drop support.
- Managed client state using Zustand across multiple feature domains and built report generation workflows with PDF and XLSX export.

Frontend Developer | Octo Technologies (Contract) — Oct 2025 - Jan 2026

- Developed and maintained features in a React Native mobile app using functional components and hooks.
- Implemented navigation logic and user-based conditional flows.
- Integrated REST APIs with asynchronous data fetching and error handling.
- Contributed to a shared codebase, debugging and adapting to evolving requirements.

Software Engineering Intern | iSpace Technologies — Jan 2025 - May 2025

- Collaborated with developers and AI tools to improve feature accuracy and user experience.
- Conducted web research and data validation to enhance implementation correctness.
- Designed structured prompts for internal automation tools, improving efficiency of data generation.
- Partnered with UI/UX designers to refine workflows for more human-centered results.

PROJECTS

CONNUIT AI WORKFLOW AUTOMATION PLATFORM | LINK

- Built a full-stack workflow platform (React, TypeScript, Express, MongoDB, BullMQ/Redis) where users chain HTTP, AI, Slack, Discord, Telegram and Gmail steps on a pan-and-zoom canvas, run by schedule, webhook or Telegram message.
- Designed a graph execution engine (topological sort, loop detection) with AI-decided If branches, fan-out to three steps, explicit merges and skipped-path tracking; runs retry automatically on a Redis job queue.
- Enforced daily AI quotas per user and platform-wide with a single atomic MongoDB upsert: 20 concurrent runs against a limit of 5 admitted exactly 5. Limit failures are not retried.
- Secured integrations with AES-256-GCM encrypted credentials, Google OAuth (least-privilege gmail.send), JWT auth and per-workflow HMAC webhook secrets.
- Deployed the API on AWS EC2 (systemd, Caddy auto-HTTPS) and the frontend on Vercel.

GLYCOTRAC DIABETES MEDICATION TRACKER | LINK

- Built a full-stack diabetes medication tracker (Next.js, TypeScript, Supabase) with dose logging, refill alerts, readings, symptom log and a printable doctor summary.
- Enforced per-user data privacy with Postgres row-level security and atomic database functions for dose logging.
- Implemented scheduled Web Push reminders and missed-dose alerts (service worker, pg_cron); built an accessible, mobile-first UI deployed on Vercel.

NESTFIND REAL ESTATE PLATFORM | LINK

- Built a full-stack real estate listing platform using Next.js, Express, and MongoDB with JWT-based authentication and role-based access control (user/moderator/admin).
- Implemented real-time buyer-seller messaging with Socket.IO, including per-conversation unread tracking and room-based delivery.
- Designed a listing verification workflow with moderator review, plus rate-limited auth endpoints and Cloudinary-backed image uploads.
- Wrote an integration test suite (Vitest, Supertest, mongodb-memory-server) and deployed the app across Render and Vercel.

OPENCLAW SELF-HOSTED AI AGENT

- Deployed a self-hosted AI agent gateway on AWS EC2, configuring systemd services, process persistence (linger), and firewall/security-group rules to keep the service running independently of any active session.
- Integrated Telegram as a chat channel with token-based auth and a pairing/approval flow for access control.
- Secured remote access via SSH tunneling, keeping the admin dashboard restricted to loopback rather than exposed publicly.

CERTIFICATIONS | LINK

Meta Frontend Web Development Certification

- Built responsive web applications with React, utilizing components, state management, routing and documentation best practices.

Google CyberSecurity Certification

- Applied cybersecurity principles including threat detection, network defense, and risk assessment.